Bach Nguyen

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Education

Ontario Tech University Sep. 2023 – Apr. 2027 (Expected)
BSc (Honours), Integrated Mathematics & Computer Science (Co-op) *Oshawa, ON*

- GPA: 3.77/4.3 | President's List (3×), Dean's List
- Relevant Coursework: Database Systems, Data Mining, Scientific Data Analysis, Statistics & Probability

Technical Skills

Languages: Python, C++, Java, SQL, Bash
Machine Learning: PyTorch, vision transformers, continual learning, scikit-learn, NumPy, pandas
Data & ML Systems: PySpark, Kafka, Delta Lake, Databricks, Docker, GitHub Actions
Backend & Cloud: FastAPI, REST APIs, Azure Container Apps, Azure Blob Storage, Cosmos DB, Git

Research & Professional Experience

Undergraduate Thesis Researcher, Visual Computing Lab Sep. 2026 – Present
Ontario Tech University, supervised by Dr. Faisal Qureshi *Oshawa, ON*

- Implemented a 10M-parameter **CascadedViT** variant in **PyTorch** to investigate parameter-efficient continual visual learning
- Investigating **SEMA** expert-routing variants to adapt learned representations across tasks, collaborating with a CascadedViT paper co-author

IT Intern May 2025 – Aug. 2025
Canadian Mental Health Association *Oshawa, ON*

- Built and deployed scheduling workflows with **Power Apps** and **Power Automate**, replacing manual bookings and eliminating recurring scheduling conflicts
- Gathered user requirements, validated workflow outputs, and documented the solution for ongoing support; provided IT assistance to 100+ staff

Teaching Assistant, Prep. Math for Engineers (MATH-0900U) Sep. 2026 – Present
Ontario Tech University *Oshawa, ON*

- Lead weekly engineering mathematics tutorials and assess student coursework and examinations

Projects

MarketGuard: Streaming Data & Validation Pipeline | *Python, Kafka, PySpark, Delta Lake* | GitHub 2026

- Built a local **Kafka** and **Spark Structured Streaming** pipeline to validate, deduplicate, and aggregate market trades into Bronze/Silver/Gold layers
- Implemented replay-safe **Delta Lake** operations and tests for duplicate events, checkpoint recovery, and numerical accuracy
- Developed a separate **Azure Blob Storage** ingestion path and published validated analytics snapshots for a dashboard

Ask Bach: Deployed Retrieval-Augmented AI Assistant | *Python, FastAPI, Azure, Docker* | GitHub 2026

- Independently built and deployed a **FastAPI** service on Azure Container Apps with hybrid BM25 and vector retrieval
- Implemented schema-validated LLM outputs with verified source IDs; developed **Cosmos DB** document and vector-search integration

Leadership

Events Coordinator Sep. 2025 – Present
Ontario Tech Computer Science Club *Oshawa, ON*

- Co-organize an annual 250+ participant hackathon: logistics, sponsor relations, and event-day operations